

Seungpil Choi

Curriculum Vitae

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Phone number: +82-10-5478-5334

Email: csp213@hufs.ac.kr

EDUCATION

2019 – 2024 Bachelor of Computer and Electronic System Engineering
Hankuk University of Foreign Studies, South Korea

GPA: 3.7/4.0 (**Major GPA:** 3.91/4.0)

Relevant Coursework: Computer Vision, Machine Learning, Data Mining, Probability and Statistics, Linear Algebra, Python programming, Algorithm, Data Structure, SW Research Project and Practice, Medical Image Processing & Laboratory

RESEARCH INTERESTS

- Signal Processing
- Computer Vision
- Medical Image Analysis
- Deep Reinforcement Learning

PUBLICATIONS

International Journal

Seungpil Choi[†], Seoyeon Jang[†], Sunghee Jung, Heon Jae Cho, Byunghwan Jeon*
Deep Reinforcement Learning for Efficient Registration between Intraoral-scan Meshes and CT Images, *Pattern Recognition*, 2025 (**IF 7.5**), <https://doi.org/10.1016/j.patcog.2025.111502>

- Citation
 - Alsheghri, A., Ladini, Y., Hosseinimanesh, G., Chafi, I., Keren, J., Cheriet, F. and Guibault, F., 2024. Adaptive point learning with uncertainty quantification to generate margin lines on prepared teeth. *Applied Sciences*, 14(20), p.9486.
 - Zhang, Z., Zhou, J., Yao, J., Liu, L., Li, J., Li, L. and Wu, X., 2025. Learning label-specific features for multi-dimensional classification. *Pattern Recognition*, p.112365.
 - Ramakoteswararao, S., Palanivel, S. and Murthy, G.L.N., 2025. An Intelligent Multimodal Medical Image Registration using Hybrid Meta-Heuristic Optimization with Transformer-based Residual UNet. *Pattern Recognition*, p.112708.
 - Pei, Y., Zhang, H., Gao, C., Müller, M., Zhu, M., Sheng, H., Chen, Z., Lin, L. and Zhu, H., 2025. Double Check My Desired -Return: Transformer with Target Alignment for Offline Reinforcement Learning. *arXiv preprint arXiv:2508.16420*.
 - Slimani, K., Achard, C. and Tamadazte, B., 2025. iMatcher: Improve matching in point cloud registration via local-to-global geometric consistency learning. *arXiv preprint arXiv:2509.08982*.

Seoyeon Jang[†], **Seungpil Choi[†]**, Byunghwan Jeon*
Reinforcement Learning-Based Centerline Tracking with Cross-Sectional Similarity for Enhancing Mandibular Canal Segmentation, *IEEE Access*, 2025, <https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11202909>

Domestic Conference

Seungpil Choi[†], Seoyeon Jang[†], Byunghwan Jeon*
Negative Curvature-based Feature Point Extraction from 3D Oral-scanned Mesh. *35th Workshop on Image Processing and Image Understanding, Republic of Korea*, 2023

[†] co-first author

RESEARCH EXPERIENCE

Machine Intelligence Lab, Hankuk University of Foreign Studies

Aug 2025 – Present Gradient-free Adaptation to Scanning Strategy Variations via Meta-Reinforcement Learning for 3D Ultrasound Reconstruction
Role: Analyzing scanning paths, Implementation a meta-reinforcement learning framework

Oct 2023 – Mar 2025 Enhancing Mandibular Canal Segmentation Performance Using Reinforcement Learning-Based Centerline Extraction
Role: Implementing a reinforcement learning system for centerline tracking of mandibular canals, Implementing a reinforcement learning system for mental and mandibular foramen landmarks detection

Nov 2023 – Jan 2025	Smart Curling Coaching Systems (<i>Ministry of Culture, Sports and Tourism</i>) Role: Implementing a multi-agent reinforcement learning system for optimal curling strategies coaching, Implementing a curling simulator, Implementing a imitation learning, Mental and physical abnormal detection
Sep 2022 – Oct 2023	Deep Reinforcement Learning for Efficient Registration between Intraoral-scan Meshes and CT images. Role: Implementing feature points extraction from 3D meshes, Implementing reinforcement learning system for registration, Implementing a refinement method

PERSONAL RESEARCH EXPERIENCE

Oct – Dec 2023	Subcortical Segmentation in MRI Images localized by Deep Reinforcement Learning
Oct – Dec 2023	Construction Material Recommendation System Using Image and Text Embedding Based on Autoencoder
Sep – Nov 2022	Classification of cracks and potholes on roads using Open Set Recognition

TEACHING EXPERIENCE

Hankuk University of Foreign Studies		
Fall 2023	Algorithm Course	Role: Assignment creation and grading
Fall 2022	Object-Oriented Programming Course	Role: Laboratory Sections for Java, Assignment creation and grading
Minor Qualification Training for Middle and High School Teachers (Gyeonggi-do Office of Education)		
Jan 2023 – Aug 2023	Computer Programming	Role: Laboratory Sections for C++, Python
	Physical SW Engineering	Role: Laboratory Sections for Microprocessors
	Algorithm	Role: Laboratory Sections for Python
	Computer Architecture	Role: Answering questions
	Operating Systems	
	Database	
	Computer Network	
	Artificial Intelligence	

SKILLS

Programming Languages: Python, Java, C/C++
Related Framework or Library: VTK, Tensorflow, Pytorch

AWARDS / SCHOLARSHIPS

Mar 2023 – Present	Stipend for Researching in Machine Intelligence Lab
Fall 2020, 2023	Scholarship for Academic Excellence
Spring 2023	Excellent TOPCIT (Test Of Practical Competency in ICT) score (Top 10%)
Fall 2020	Merit Scholarship